

# 1 Find Your Method

Use this decision tree to navigate to the chapter or case study most relevant to your research question.

flowchart TD A["What is your research question?"] --> B["Map a research field"] A --> C["Evaluate research performance"] A --> D["Detect emerging topics or trends"] A --> E["Study collaboration patterns"] A --> F["Track open access & open science"]

B --> B1["Bibliographic coupling<br>Ch. 16"]  
B --> B2["Co-citation analysis<br>Ch. 16"]  
B --> B3["Science mapping<br>Ch. 19"]  
B --> B4["Topic modeling<br>Ch. 22"]  
B --> B5["Case Study 1:<br>CRISPR Field Review"]

C --> C1["Author metrics<br>Ch. 10, 12"]  
C --> C2["Journal metrics<br>Ch. 11"]  
C --> C3["Institutional benchmarking<br>Ch. 13"]  
C --> C4["Case Study 2:<br>Institutional Benchmark"]  
C --> C5["Case Study 3:<br>Journal Portfolio"]

D --> D1["Keyword bursts<br>Ch. 25"]  
D --> D2["Temporal analysis<br>Ch. 14"]  
D --> D3["Embedding clustering<br>Ch. 24"]  
D --> D4["Case Study 5:<br>Emerging Topics"]

E --> E1["Co-authorship networks<br>Ch. 15"]  
E --> E2["Community detection<br>Ch. 18"]  
E --> E3["Gender & equity<br>Ch. 28"]  
E --> E4["Case Study 4:<br>Gender Gap"]

F --> F1["OA indicators<br>Ch. 27"]  
F --> F2["Altmetrics<br>Ch. 26"]  
F --> F3["Reproducibility<br>Ch. 33"]

## 1.1 Quick reference table

| Research question              | Key chapters | Case study              |
|--------------------------------|--------------|-------------------------|
| Map a research field           | 16, 19, 22   | CS 1: CRISPR            |
| Evaluate an author             | 10, 12       | —                       |
| Evaluate a journal             | 11           | CS 3: Journal Portfolio |
| Benchmark an institution       | 13           | CS 2: Institutional     |
| Detect emerging topics         | 14, 24, 25   | CS 5: Emerging Topics   |
| Analyze collaboration          | 15, 18       | —                       |
| Study gender gaps              | 28           | CS 4: Gender Gap        |
| Track open access              | 27           | —                       |
| Build a dashboard              | 35, 37       | —                       |
| Create a reproducible pipeline | 34, 36       | —                       |